מועד א תשס״ה

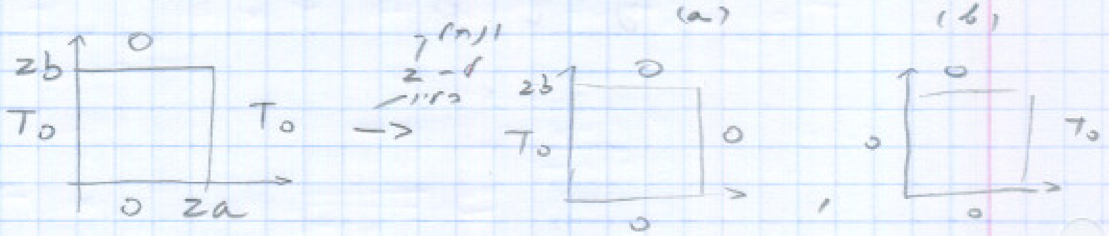
$T(-a,y) = T(a,y) = T_0$ .1

$T(x,-b) = T(x,+b) = 0$

$\Delta T = 0$

$y \to y+b$ נבצע הזזה

$x \to x+a$



a) הפתרון הכללי

$$T_a(x,y) = \sum_{n=1}^{\infty} \sin\left(\frac{\pi n y}{2b}\right)\left(A_n e^{\frac{\pi n}{2b}x} + B_n e^{-\frac{\pi n}{2b}x}\right)$$

$$0 = T_a(2a,y) \Rightarrow A_n e^{\frac{\pi n}{2b}2a} + B_n e^{-\frac{\pi n}{2b}2a} = 0$$ נדרוש

$$B_n = -A_n e^{\frac{4\pi n a}{2b}}$$

$$\Rightarrow A_n e^{\frac{\pi n}{2b}x} - A_n e^{\frac{4\pi n a}{2b} - \frac{\pi n}{2b}x} \propto \sinh\left(\frac{\pi n}{2b}(x-2a)\right)$$

$$\Rightarrow \quad T_a(x,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n}{2b}(x-2a)\right)\sin\left(\frac{\pi n y}{2b}\right)$$

$$T_0 = T(0,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n}{2b}(-2a)\right)\sin\left(\frac{\pi n y}{2b}\right)$$ [illegible]

$$A_n = \frac{2}{2b}\frac{T_0}{\sinh\left(\frac{\pi n}{2b}(-2a)\right)}\int_0^{2b}\sin\left(\frac{\pi n y}{2b}\right)dx$$

$$= \frac{1}{b}\frac{T_0}{-\sinh\left(\frac{\pi n a}{b}\right)}\left[\frac{-\cos\left(\frac{\pi n y}{2b}\right)}{\left(\frac{\pi n}{2b}\right)}\right]\Bigg|_0^{2b}$$

$$= \frac{T_0 2b}{-\pi n \sinh\left(\frac{\pi n a}{b}\right)}\left[-\cos(\pi n)+1\right] = \frac{-T_0 2b}{\pi(2k+1)\sinh\left(\frac{\pi a(2k+1)}{b}\right)}$$

$k = 0,1,2$

$$\Rightarrow T_a(x,y) = \sum_{k=0}^{\infty} \frac{2T_0 b \sinh\left(\frac{\pi(2k+1)}{2b}(2a-x)\right)}{\pi(2k+1)\sinh\left(\frac{\pi a(2k+1)}{b}\right)} \sin\left(\frac{\pi(2k+1)}{2b} y\right)$$

b) כעין בכוי ר הפתרון הכ״ל לכי עך (גרוש)

$$0 = T_b(0,y) \Rightarrow B_n = -A_n$$

$$T_b = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n x}{2b}\right) \sin\left(\frac{\pi n y}{2b}\right)$$

$$T_0 = T_b(2a,y) = \sum_{n=1}^{\infty} A_n \sinh\left(\frac{\pi n a}{b}\right) \sin\left(\frac{\pi n y}{2b}\right)$$

כאן קוצי ולכן

$$A_{2k+1} = \frac{2T_0 b}{\pi(2k+1)\sinh\left(\frac{\pi a(2k+1)}{b}\right)}$$

והפתרון

$$T_b(x,y) = \sum_{k=0}^{\infty} \frac{2T_0 b \sinh\left(\frac{\pi n x}{2b}\right)}{\pi(2k+1)\sinh\left(\frac{\pi a(2k+1)}{b}\right)} \sin\left(\frac{\pi n y}{2b}\right)$$

הפתרון הכולל יהיה סכום $T_a + T_b$ ואז הצבה

$$x \to x-a \qquad ; \qquad y \to y-b$$

$$T(x,y) = \sum_{k=0}^{\infty} \frac{2T_0 b \sin\left(\frac{\pi n(y+b)}{2b}\right)}{\pi(2k+1)\sinh\left(\frac{\pi a(2k+1)}{b}\right)} \left[\sinh\left(\frac{\pi n}{2b}(x+a)\right) + \sinh\left(\frac{\pi n}{2b}(a-x)\right)\right]$$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} + ax$$ .2

$u(x=-L,t) = u(x=L,t) = 0$

$u(x,0) = 0$

נבצע הצבה $y = x + L$

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial y^2} + a(y-L)$$

נחפש פתרון פרטי המקיים את תנאי השפה !

$$D\frac{\partial^2 u}{\partial y^2} + a(y-L) = 0$$

$$\frac{\partial^2 u}{\partial y^2} = -\frac{a}{D}(y-L) = -\frac{a}{D}y + \frac{a}{D}L$$

$$u_p(y) = -\frac{a}{D}\frac{y^3}{6} + \frac{aL}{D}\frac{y^2}{2} + Ay + B$$

$0 = u_p(0) \Rightarrow \quad B = 0$

$0 = u_p(2L) \Rightarrow -\frac{a2^3L^3}{6D} + \frac{aL2^2L^2}{2D} + A2L$

$$0 = -\frac{a4L^2}{6D} + \frac{a2L^2}{2D} + \cancel{2}A\cancel{L}$$

$$A = \frac{aL^2}{D}\left(\frac{4}{6} - 1\right) = -\frac{aL^2}{3D}$$

$$u_p(y) = \frac{a}{D}\left(-\frac{y^3}{6} + \frac{Ly^2}{2} - \frac{L^2y}{3}\right)$$

נפתור את הבעיה ההומוגנית:

$$\frac{\partial u_H}{\partial t} = \frac{\partial^2 u_H}{\partial y^2}$$

$$\Rightarrow u(y,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{\pi n y}{2L}\right) e^{-\frac{Dt\pi^2 n^2}{4L^2}}$$

כאשר יש למצוא המקדמים + הצבה ו- $y = x+L$.

3.

$$\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} = 0$$

מ. היפרבולית

$u(y=x) = \sin x$

$\left.\frac{\partial u}{\partial x}\right|_{y=x} = 1 \qquad -\infty < x < \infty$

$$\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y} + u\right) = 0$$

$$\frac{\partial u}{\partial y} + u = f(y)$$

(נפתור את הבעיה בעזרת

$u(x,y) = c(x,y)e^{-y}$

בדיקה:

$$\frac{dc}{dy}' e^{-y} - c\cancel{e^{-y}} + c\cancel{e^{-y}} = f(y)$$

$$\Rightarrow \quad \frac{dc}{dy} = f(y)e^{y}$$

$$c = \int f(y)e^{y}dy = \tilde{f}(y) + \tilde{g}(x)$$

$$\Rightarrow \quad u(x,y) = (\tilde{f}(y) + \tilde{g}(x))e^{-y}$$

$$u(y=x) = (\tilde{f}(x) + \tilde{g}(x))e^{-x} = \sin x$$

$$\left.\frac{\partial u}{\partial x}\right|_{y=x} = \left. g'(x)e^{-y}\right|_{y=x} = 1$$

$$\Rightarrow \quad \tilde{g}(x) = e^{x} + d$$

$$\Rightarrow \quad (\tilde{f}(y) + e^{y} + d)e^{-y} = \sin y$$

$$= \tilde{f}(y) = e^{y}\sin y - d - e^{y}$$

$$\Rightarrow \quad u(x,y) = (e^{y}\sin y \cancel{-d}\,{}^{y} e^{x})e^{-y}$$

$$u(x,y) = (e^{y}(\sin y - 1) + e^{x})e^{-y}$$

4.

$$\hat{L}y(x) = x^2 y''(x) + 2x y'(x)$$

$$1 < x < 2$$

$$y(1) = y(2) = 0$$

נמצא את פונ' גרין של האופרטור. (הגדרתה היא המשוואה:

$$x^2 G''(x,x') + 2x G'(x,x') = \delta(x-x')$$

$$\frac{d}{dx}\left(x^2 \frac{dG(x,x')}{dx}\right) = \delta(x-x')$$

עבור $x \neq x'$

$$\frac{d}{dx}\left(x^2 \frac{dG(x,x')}{dx}\right) = 0$$

$$x^2 \frac{dG(x,x')}{dx} = C_1(x')$$

$$\frac{dG(x,x')}{dx} = \frac{C_1(x')}{x^2}$$

$$\frac{G(x,x')}{\cancel{d}} = \frac{C_1(x')}{x} + C_2(x')$$

נשתמש בתנאי השפה

$$0 = G(1,x') \Rightarrow -C_1(x') = +C_2(x')$$

$$0 = G(2,x') \Rightarrow \frac{C_3(x')}{2} + C_2(x') = 0$$

$$C_2 = -\frac{C_3}{2}$$

$$\Rightarrow G(x,x') = \begin{cases} C_1(x')\underbrace{\left(\frac{1}{x} - 1\right)}_{y_1(x)} & 1 \leq x \leq x' \\ C_2(x')\underbrace{\left(\frac{1}{x} - \frac{1}{2}\right)}_{y_2(x)} & \end{cases}$$

$$\left.\frac{dG_>}{dx} - \frac{dG_<}{dx}\right|_{x'} = \frac{1}{p(x')}$$

+ רציפות של הפונ' ב $x = x'$

מתנאי הרציפות והנגזרת נקבל

$$c_1(x')y_1(x') - c_2(x')y_2(x') = 0$$

$$c_1(x)y'_1(x') - c_2(x')y'_2(x') = -\frac{1}{p(x')}$$

$$\Rightarrow \begin{pmatrix} y_1(x') & -y_2(x') \\ y_1'(x) & -y_2'(x) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{p(x')} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \frac{1}{x'}-1 & \frac{1}{2}-\frac{1}{x'} \\ -\frac{1}{x'^2} & \frac{1}{x'^2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{1}{x'^2} \end{pmatrix}$$

$$\Delta = \frac{1}{x'^2}\left(\frac{1}{x'}-1\right) + \frac{1}{x'^2}\left(\frac{1}{2}-\frac{1}{x'}\right) = \frac{1}{x'^2}\left(-\frac{1}{2}\right)$$

$$\Delta c_1 = \frac{1}{x'^2}\left(\frac{1}{2}-\frac{1}{x'}\right)$$

$$c_1 = \frac{\Delta c_1}{\Delta} = \frac{\frac{1}{2}-\frac{1}{x'}}{-\frac{1}{2}} = 2\left(\frac{1}{x'}-\frac{1}{2}\right)$$

$$\Delta c_2 = \frac{1}{x'^2}\left(1-\frac{1}{x'}\right)$$

$$c_2 = \frac{\Delta c_2}{\Delta} = 2\left(\frac{1}{x'}-1\right)$$

$$G(x,x') = \begin{cases} 2\left(\frac{1}{x'}-\frac{1}{2}\right)\left(\frac{1}{x}-1\right) & 1 \le x \le \overset{\text{[illegible]}}{x'} \\ 2\left(\frac{1}{x}-\frac{1}{2}\right)\left(\frac{1}{x'}-1\right) & x' \le x \le 2 \end{cases}$$

$$\varphi(x) = \lambda \iiint_{G_1} \frac{1+|x'|}{1+|x|} \varphi(x')\,dx'$$ .5

$$= \frac{\lambda}{1+r} 4\pi \int_0^1 (1+r')\,r'^2\,dr\ \varphi(r')$$

$$\varphi = \frac{4\pi\lambda}{1+r} \frac{I\,r'^3}{3} + \frac{r'^4}{4}\Big|_0^1 = \frac{1}{4} \quad 4+\ \frac{7}{12}$$

$$\Rightarrow I = \frac{4\pi}{1+r} \int_0^1 \frac{\lambda I \cancel{(1+r')}}{\cancel{1+r'}}\, r'^2\,dr'$$

$$= \frac{4\pi\lambda I}{3(1+r)}$$

$$1 = \frac{4\pi}{3} \frac{\lambda}{1+r}$$

$$\varphi = \frac{\lambda I}{1+r}$$

$$I = \int_0^1 (1+r')\,r'^2\, \frac{4\pi\lambda I}{1+r'}\,dr'$$

$$= \frac{4\pi}{3} \lambda I$$

$$\lambda = \frac{3}{4\pi}$$

$$\varphi = \frac{3I}{1+r}$$ I - קבוע שרירותי